

Are You Getting What You Pay For? Auditing Model Substitution in LLM APIs

The trust gap in commercial LLM
APIs\nReference:

<https://arxiv.org/abs/2504.04715>

The Model Substitution Problem: Economic Incentives Drive Covert Swaps

- Providers may replace advertised models with cheaper alternatives to cut costs.
- Hypothesis Testing Framework: Can an auditor distinguish $P_{\text{spec}}(y|x)$ from $P_{\text{actual}}(y|x)$?
- Four main substitution types:
 - 1. Smaller models (e.g., 8B instead of 70B)
 - 2. Quantized variants (FP8/INT8 instead of FP16/FP32)
 - 3. Fine-tuned or updated versions
 - 4. Entirely different model families

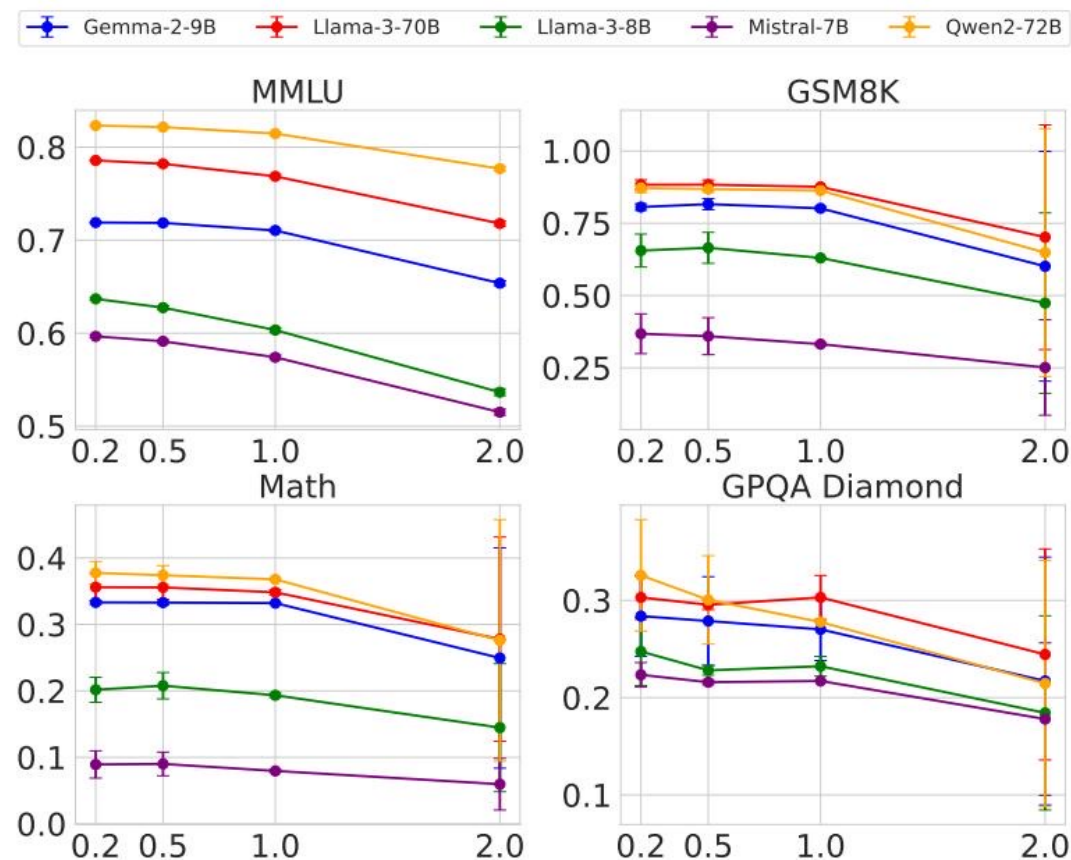
FP8 Quantization Preserves Most Accuracy — Making Substitution

Performance drops are small (often within error bars), motivating why detection is difficult.

Hard to Detect

Model	MMLU	GSM8K	MATH	GPQA Diamond
Llama-3-8B-Instruct	62.69 ± 0.18	61.14 ± 3.47	20.65 ± 3.43	22.62 ± 0.22
Llama-3-8B-Instruct-FP8	62.43 ± 0.26	60.90 ± 4.10	14.91 ± 2.49	20.14 ± 0.24
Llama-3-70B-Instruct	78.05 ± 0.08	88.06 ± 1.44	35.69 ± 1.33	29.60 ± 0.51
Llama-3-70B-Instruct-FP8	77.88 ± 0.13	87.35 ± 1.24	35.75 ± 1.16	33.12 ± 0.30
Gemma-2-9b-it	71.86 ± 0.08	81.80 ± 1.35	33.41 ± 0.28	28.64 ± 2.97
Gemma-2-9b-it-FP8	71.92 ± 0.11	79.41 ± 1.14	32.53 ± 0.34	27.81 ± 3.22
Qwen2-72B-Instruct	82.18 ± 0.08	86.72 ± 1.00	37.39 ± 1.41	29.93 ± 2.71
Qwen2-72B-Instruct-FP8	81.98 ± 0.08	86.82 ± 0.97	37.67 ± 1.39	31.08 ± 1.96
Mistral-7B-Instruct-v0.3	59.15 ± 0.10	35.90 ± 4.54	8.94 ± 1.24	21.60 ± 0.17
Mistral-7B-Instruct-v0.3-FP8	58.77 ± 0.13	32.20 ± 4.02	7.68 ± 1.23	22.72 ± 0.19

Higher Temperature Amplifies Performance Gaps but Increases



At higher temperatures (e.g., 2.0), performance degrades across all tasks, and the gap between original and substitute narrow

Log Probability Fingerprinting Fails Due to Inference Nondeterminism

Model	BERT Acc	T5 Acc	GPT2 Acc	LLM2
Llama3-70B-Instruct-FP8	50.55	50.10	50.45	49
Llama3-70B-Instruct-INT8	51.60	49.90	51.30	50
Gemma2-9b-it-FP8	49.95	50.30	51.20	49
Gemma2-9b-it-INT8	49.00	49.70	51.65	49
Mistral-7b-v3-Instruct-FP8	50.55	49.75	48.70	48
Mistral-7b-v3-Instruct-INT8	49.70	50.75	50.50	51
Qwen2-72B-Instruct-FP8	50.05	50.25	48.75	49
Qwen2-72B-Instruct-INT8	50.75	50.55	49.45	50

The same model produces varying log probabilities across different software versions and hardware. Production-level nondete

Randomized Routing at Low Rates Makes Statistical Detection Query- Prohibitive

- Providers can actively evade audits using adversarial countermeasures like Randomized Substitution.
- Attack Mechanism: Route only a fraction 'p' of queries to the cheaper model.
- Statistical Impact: As 'p' approaches 0, the mixed distribution tightly converges to the specified model's distribution.
- Distribution Formula: $P_{\text{observed}}(y|x) = (1 - p) * P_{\text{spec}}(y|x) + p * P_{\text{actual}}(y|x)$

Software-Only Auditing Methods Are Fundamentally Unreliable

Method Type	Weakness	Practical Limitation
Text-based Statistical Tests (e.g., MMD)	Needs too many queries; fails against subtle FP8 substitutions.	Impractical query budgets required.
Log-Probability Fingerprinting	Defeated by production inference nondeterminism.	Cannot distinguish software/hardware variation from substitution.
Adversarial Countermeasures	Benchmark evasion and randomized routing undermine detection.	Providers control the black box and can route queries dynamically.

Trusted Execution Environments

Provide Provable Cryptographic

Model Integrity

- Hardware-secured enclaves (TEEs) provide cryptographic attestation of the code and data being executed.
- Guarantees that the correct model weights are loaded and the correct inference code is running.
- Provable, efficient, and robust — in stark contrast to software-only methods.
- Unlike Zero-Knowledge Proofs (ZKPs), TEEs have only modest performance overhead.

Key Takeaways: Close the Verification Gap with Hardware Attestation

- 1. Model substitution is economically motivated and practically undetectable via software.
- 2. FP8 quantization barely affects benchmarks, masking cost-cutting.
- 3. Inference nondeterminism kills log-probability fingerprinting.
- 4. Randomized substitution actively defeats statistical tests.
- 5. TEEs are the only currently viable path to